

AUTONOMOUS And Connected Car Technologies Are Making The VEHICLES Safe, Intelligent, And Efficient

Alexander Klotz

head of Technical Center India, Continental Automotive



With a focus on safety and economic growth, connected and autonomous vehicles are the future of mobility in smart cities. Alexander Klotz, head of Technical Center India, Continental Automotive, explains what needs to be done to achieve various capabilities of such vehicles and their benefits, in an interaction with Ayushee Sharma

Q. What features are necessary to term a vehicle as holistically connected?

A. Holistic connectivity enables embedded systems inside the vehicle (for example, high-performance computing, gateways, or sensors) and systems outside the vehicle (such as edge computing, cloud) interact more efficiently to achieve advanced levels of autonomous driving capabilities. However, to term a vehicle as holistically connected, wireless updates for the entire vehicle electronics is extremely crucial.

Q. What are the benefits of connected electric cars?

A. Irrespective of whether the vehicle is electric or not, connectivity enables the vehicle to receive and send data in real-time. With holistic connectivity, to keep this digital network up-to-date or to install new functions, it is possible to update while on the go.

Similarly, connectivity opens the scope for predictive maintenance, where the vehicle gets scanned for any possible breakdown, or perhaps loss in tire pressure. The driver receives this information on-board and through suggestion, based on GPS, the driver gets information about the nearest service station. In a typical scenario, electric vehicle (EV) range anxiety can be eased through real-time range calculation and suggestions on an optimally located charging station.

Connectivity also makes in-vehicle infotainment smarter and more flexible. With the cloud terminal, the system is always up-to-date and new functions can be added at any time, allowing vehicle manufacturers to roll out the applications across different models and vehicle classes. This also helps in making the vehicle safer.

Many autonomous driving features have become a reality only because of connectivity. For example, Continental's Cellular V2X (C-V2X) communication allows an exchange of such time-sensitive and safety-critical information as warnings of potentially hazardous situations.

Q. What are the key considerations in designing connected vehicles?

A. It is a whole new game from an interior perspective where infotainment systems, cameras, a head-up display, and other integrated systems are collectively referred to as vehicle cockpit.

The car of the future may well sport large interactive and integrated information display surfaces running from pillar to pillar, what we refer today as Integrated Interior Platform (IIP). Another vital consideration is the mounting of sensors that help in monitoring driver behaviour, and designing will play a key role here, too.

While considering the exteriors, vehicles will increasingly be equipped with cameras, radars, lidars, and other proximity sensors. Cameras aiding in rear viewing, 360-degree surround view for blind-spot detection, and under the hood viewing.

Q. What wireless technologies enable intra- and inter-vehicle communications?

A. Near Field Communications (NFC) has been used in various inter-vehicle communications. It is a set of short-range wireless technologies, typically requiring a distance of ten centimetres or less, for two devices such as smartphones to establish communication. Continental has also been using NFC for smart access and door-handle sensor.

Ultra-Wideband is another technology that complements the existing short-distance radio standards in keyless vehicle access. It helps to avoid unauthorised access via relay (man-in-the-middle) attacks and therefore makes keyless vehicle access more secure.

Dedicated Short Range Communication (DSRC) allows vehicles to directly communicate with each other through over-the-air messages compliant to the IEEE 802.11p standard. Apart from DSRC, cellular V2X (Vehicle to Everything) communication has a strong potential of becoming a key enabler for automated driving and intelligent mobility.

Q. What are the challenges involved in achieving the transition from partially to fully autonomous vehicles?

A. What we need today is a more reliable artificial intelligence (AI) that offers accurate machine learning (ML) techniques to enable the system to sense the road environment and take intelligent decisions. Given that all the collected data needs to be combined, processed, and turned into a machine-friendly picture of the environment, and further make decisions to move through the environment, it calls for high-performance computing power within the vehicle.

Currently, there is no standard operating system (OS) for vehicles. This creates multiple ecosystems that might eventually cause trouble to the user while migrating cars or being connected across different systems. Hackers might be motivated to hack into the vehicle's system to steal vital information of the passenger or meddle with the functionalities of the car.

Besides, the sensors like lidar, radar, and camera are getting higher in numbers and increasing in complexities. **EFY**